

K Means Clustering

K-means clustering is an unsupervised ML algorithm, used to group data points into K clusters based on the similarity of the data.

③ Steps :-

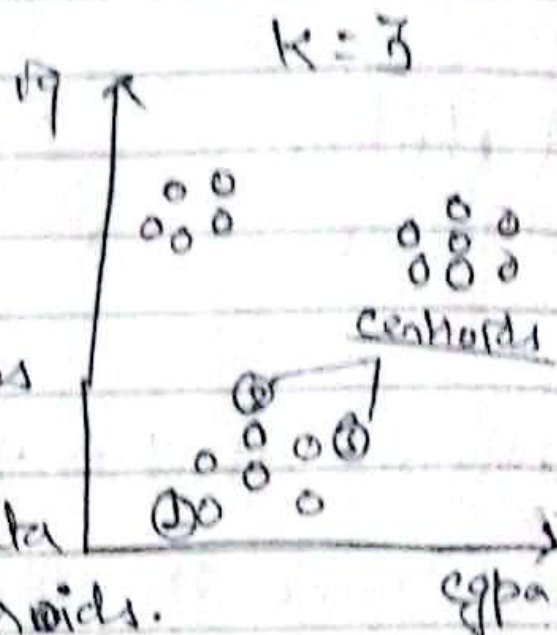
1) Select the no. of clusters

2) Randomly picks K data points as initial centroids.

Centroid is basically the center of the cluster. Centroid is mean.

$$\text{Centroid} = \frac{1}{n_k} \sum_{i=1}^n x_i$$

3) Assign each point to the cluster. Measure euclidean distance of each point to the cluster centroid and choose the minimum distance cluster.



4) Move the centroid

5) Check the previous centroid is same as new centroid? If yes, finish the algorithm. If no, repeat step 2 after moving the centroid.

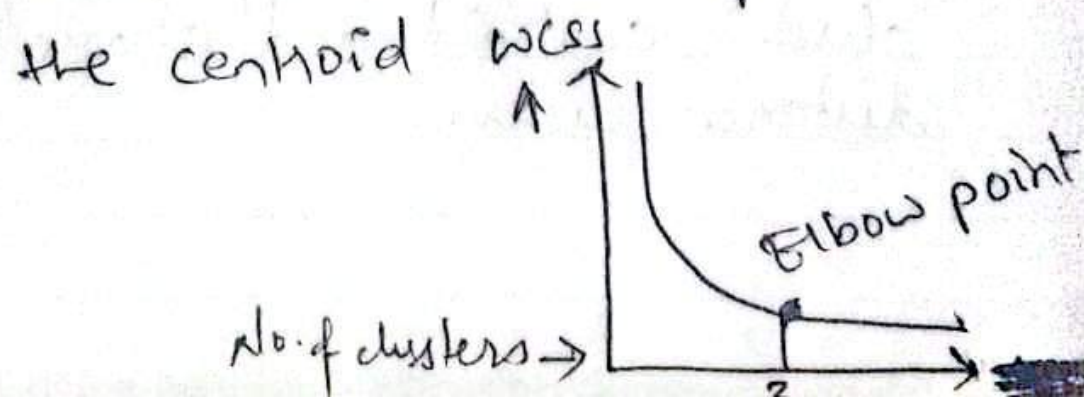
$$\text{Euclidean distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Elbow Method $\Rightarrow$ It is a technique used to determine the optimal no. of clusters in K-means clustering.

For Elbow method, we need to calculate the WCSS $\rightarrow$ within cluster sum of squares

$$WCSS = \sum_{i=1}^n d_i^2$$

where, d = Distance between a point and the centroid



How it works?

- 1) Run K-means for diff. values of K.
- 2) For each K, compute WCSS.
- 3) Plot K on X-axis & WCSS on Y axis
- 4) Look for an elbow point.

DBSCAN Clustering

Density Based Spatial Clustering of Applications with Noise.

DBSCAN is a type of unsupervised ML that groups points based on density.

The most disadvantage of K-means was

- 1) Must choose K at the beginning
- 2) Struggles with non-spherical clusters.

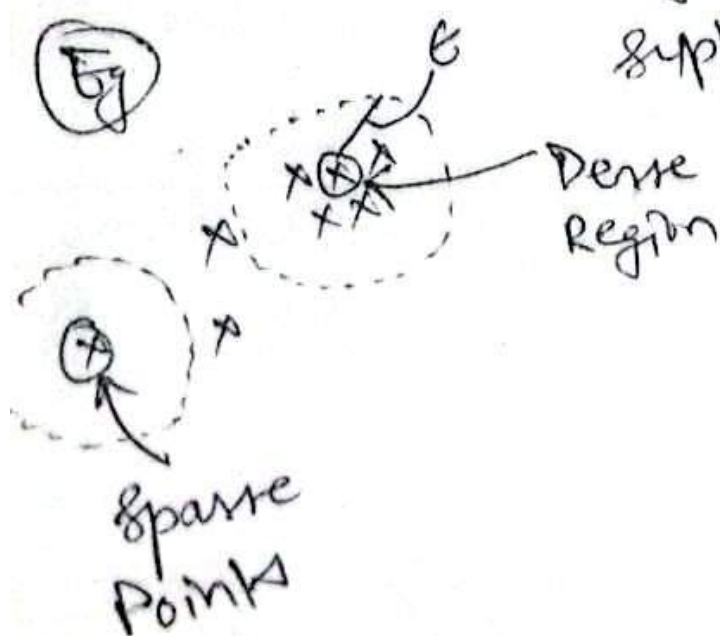
This disadvantage is solved by DBSCAN.

DBSCAN uses 2 main parameters:-

- 1) ϵ (epsilon) $\rightarrow$ Radius of Neighborhood
- 2) minPts $\rightarrow$ min. no. of points required to form a dense region

(Eg)

Suppose, $\epsilon = 1$ unit
minPts = 3

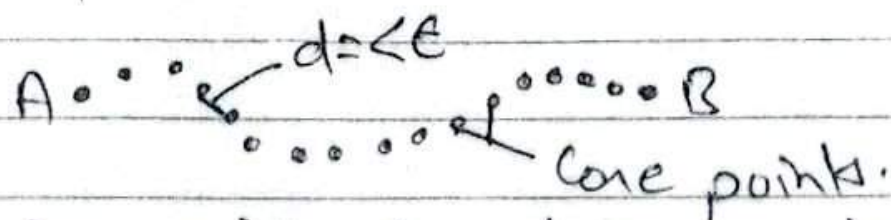


⑩ Types of Points

- 1) Core Point $\rightarrow$ Has $\geq \text{MinPts}$ in its ϵ Neighborhood
- 2) Border Point $\rightarrow$ Has $< \text{MinPts}$ but has at least one core point in its ϵ Neighborhood
- 3) Noise Points $\rightarrow$ Neither core nor border.

⑪ Density Connected Points

$\Rightarrow$ Two points A and B are called density connected points if they are connected with one or more core points and the distance between two consecutive core points is less than ϵ .



If two points A and B are density connected, we can put both points in same cluster.

Algorithm

- 1) Set the value of ϵ and MinPts
- 2) Identify all the points as either core points, or border points or noise points.
- 3) For all the unclustered core points,
 - 3)a) Create a new cluster
 - 3)b) Add all the unclustered points and density connected points to the current points to the cluster.
- 4) For each unclustered border point, assign it to the cluster of nearest core point.
- 5) Leave all the noise points as it is.

Hierarchical clustering

Hierarchical clustering is a type of unsupervised learning that builds a hierarchy (tree-like structure) of clusters.

The result of hierarchical clustering is shown in dendograms.

Types of hierarchical clustering :-

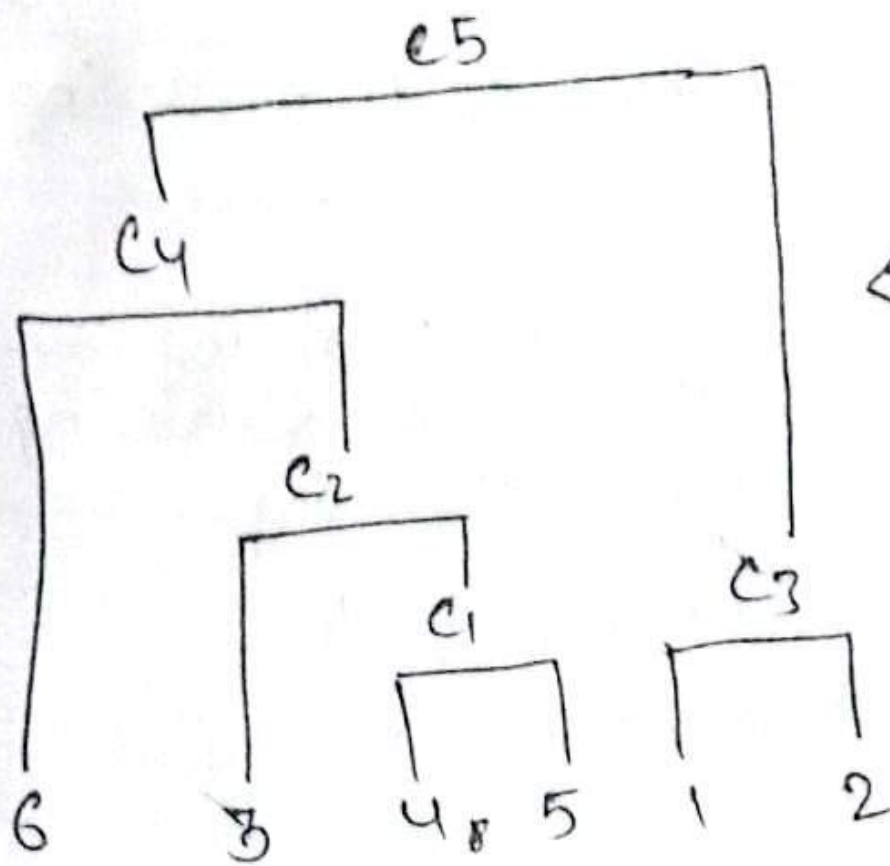
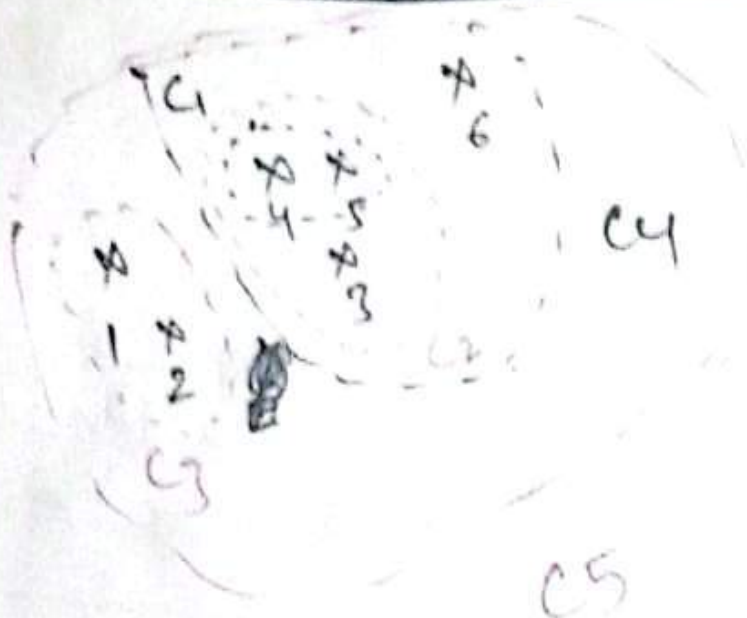
- 1) Agglomerative Hierarchical clustering
- 2) Divisive Hierarchical clustering

1) Agglomerative (Bottom-up)
→ Most common

Steps

- 1) Treat each point as a cluster
- 2) Compute distance between every pair of cluster
- 3) Merge the closest cluster
- 4) Update the distance matrix
- 5) Repeat steps until one cluster remains

∴ Key idea → start small, merge until last



← The figure is dendrogram.

At the end, you will get only one cluster in agglomerative clustering.

2) Divisive (Top-Down) - Rare

Steps

- 1) Start with one cluster containing all data
- 2) select a cluster to split
- 3) split into two small clusters (based on dissimilarity)
- 4) continue splitting until all points are separate.

∴ Key idea → start large, split until small

⊗ Distance Metrics Used

- Euclidean Distance
- Manhattan Distance

⊗ Note:- Agglomerative is mostly used than divisive because merging is easier than breaking down.

Linkage Methods

1) Single Linkage

Distance = min distance between any two points in the clusters.

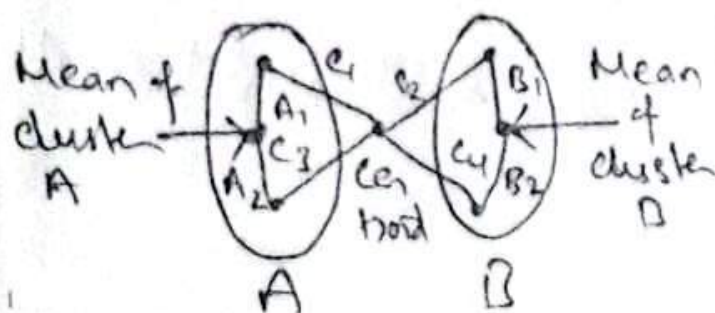
2) Complete Linkage

Distance = Max distance between two points.

3) Average Linkage

Distance = Avg distance between all pairs of points.

4) Ward's Method (Most Common & Best)



steps

- 1) compute centroid of all points
- 2) compute distance from centroid to each point
- 3) compute mean of each cluster

a) Apply formula,

$$\text{distance} = C_1^2 + C_2^2 + C_3^2 + C_4^2 - A_1^2 - A_2^2 - B_1^2 - B_2^2$$

It is the default method in sklearn.

PCA

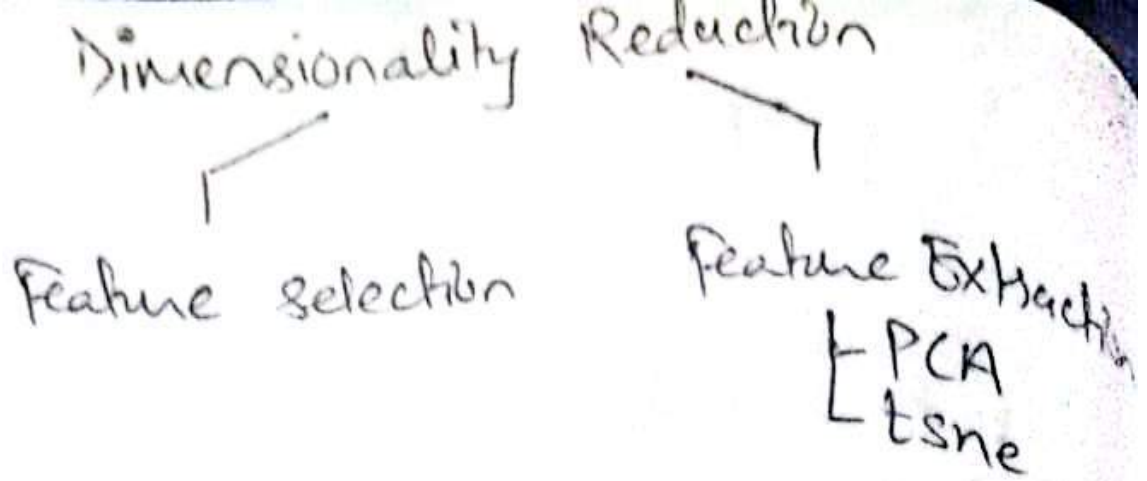
PCA stands for Principle Component Analysis.

PCA is a dimensionality reduction technique used to reduce the no. of features while keeping the max important information (variance).

⊗ Curse of Dimensionality

- curse of dimensionality refers to the problems that arise when your dataset has too many features.
- It may increase overfitting.
- Higher dimension means more computation that slows training and inference.

⊗ COD refers to the difficulties that arise when data has too many features causing sparsity, overfitting and high computational cost.



Dimensionality Reduction is the process of reducing the no. of input features while keeping as much useful information as possible.

Why Dimensionality Reduction

- Reducing overfitting
- Working with high dimensional data
- Improving training speed.

Types

1) Feature selection

→ choose the best original features.
don't create new features.

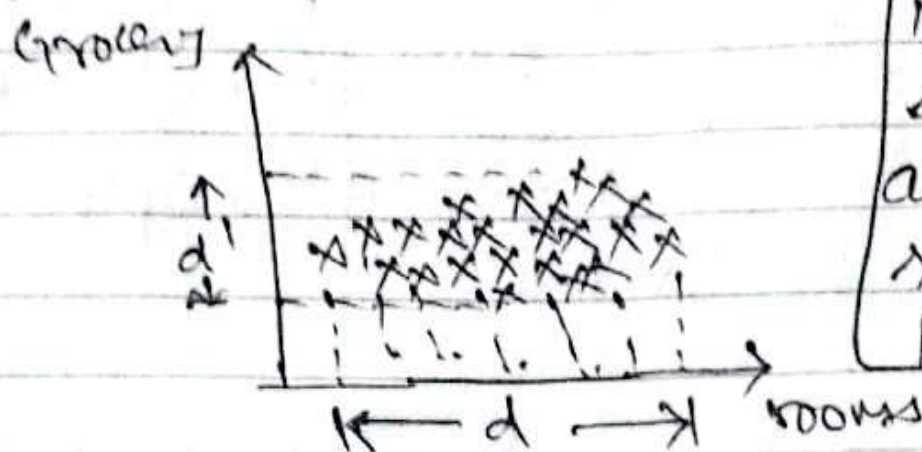
2) Feature extraction

→ Create new features from the old ones.

PCA, tSne are the feature extraction techniques.

(*) PCA Geometric Intuition

If the data is like this

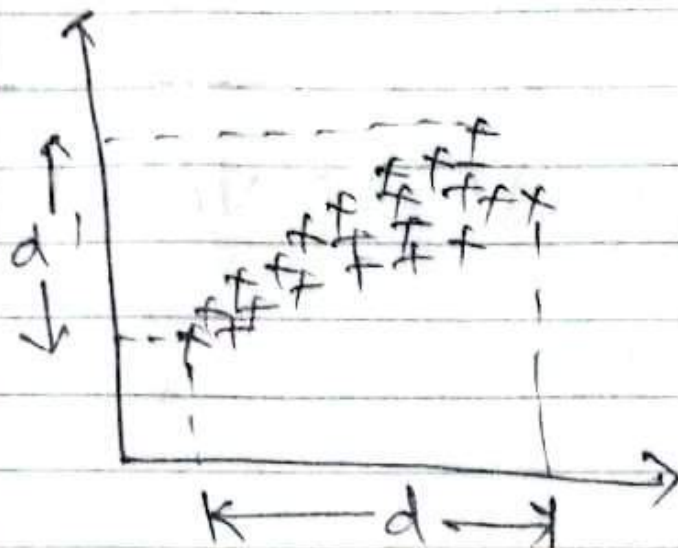


No. of grocery shops & rooms are not related. to price of flat

Calculate the data spread in x-axis and y-axis, and choose the highest spread.

Here, it is clear that $d > d'$, so we choose d over d' .

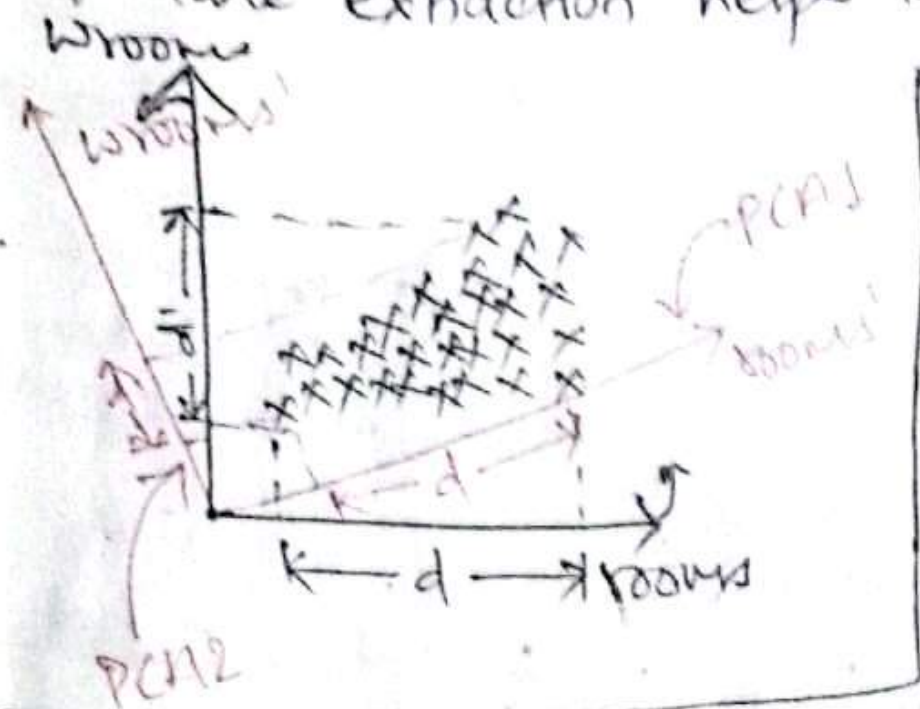
This is feature selection.



No. of rooms & no. of washrooms are related to determine the price of flat.

we can see that $d \approx d'$. So, which feature to select?

Feature extraction helps in this case.



PCA rotates the coordinates and the new coordinates formed are called Principle Components.

You get that much Principle Components as much you have features in original dataset.

You choose PC1 because variance is high in PC1 in comparison to PC2.

⊗ Best PC is the component that captures max variance.

$$\text{Var}(PC_1) > \text{Var}(PC_2) > \text{Var}(PC_3)$$

Cost fn $\rightarrow$ Variance.

⊗ EigenVectors and EigenValues

⇒ For a sq. matrix A , an eigen vector is a special vector whose direction do not change when you apply the matrix transformation.

$$AV = \lambda V$$

where,

A = square matrix

V = eigen vector

λ = eigen value (amt of stretch and shrink).

Simply,

eigenVector → A direction that remains unchanged after transformation

EigenValue → How much that direction is stretched ($\lambda > 1$) or shrunk ($0 < \lambda < 1$)

tSNE

stands for t-Distributed Stochastic Neighbor Embedding

A non-linear dimensionality reduction and visualization technique mainly used to convert high dimensional data to 2D or 3D plots while preserving local structure.

tSNE helps you visualize high dimensional data by placing similar points close together and dissimilar points far apart in 2D or 3D.

It is a visualization tool, not a clustering algorithm.

1) Convert all distances to probability values.

$$p_{ij} = \frac{e^{-|x_i - x_j|^2 / 2\sigma_i^2}}{\sum_{k \neq i} e^{-|x_i - x_k|^2 / 2\sigma_i^2}}$$

$$p_{ij}$$

$$= \frac{e^{-|x_i - x_j|^2 / 2\sigma_i^2}}{\sum_{k \neq i} e^{-|x_i - x_k|^2 / 2\sigma_i^2}}$$

⇒ Asymmetric distribution

⇒ High dimension

σ_i = Bandwidth chosen using perplexity

Each point has its own σ_i

By assuming all variances are same across the distribution, the new point in some other plane would be

$$q_{j|i} = \frac{e^{-|y_i - y_j|^2}}{\sum_{k \neq i} e^{-|y_i - y_k|^2}}$$

} Symmetric distribution
} lower dimension

You have to make the distribution symmetric in tSNE.

⊗ Minimize KL Divergence

→ It tries to make high-D prob $\approx$ low-D prob by minimizing the cost function:

KL Divergence = How different the two probability distributions are

Result: Similar points = Close
Dissimilar points = Far

$$KL(P||Q) = \sum_i \sum_j P_{ij} \log \left(\frac{P_{ij}}{Q_{ij}} \right)$$

we have to minimize this.

P_{ij} larger than Q_{ij} $\rightarrow$ high cost
means x_i, x_j are neighbors in h-D but
far apart in small-D.

P_{ij} smaller than Q_{ij} $\rightarrow$ low cost
means, x_i and x_j are far apart in high-D
but neighbors in small-D.

$$Q_{ij} = \frac{\frac{1}{1+|y_i - y_j|^2}}{\sum_{k \neq i} \frac{1}{1+|y_i - y_k|^2}}$$

$\Rightarrow$ Symmetric
 $\Rightarrow$ uses
t-distribution